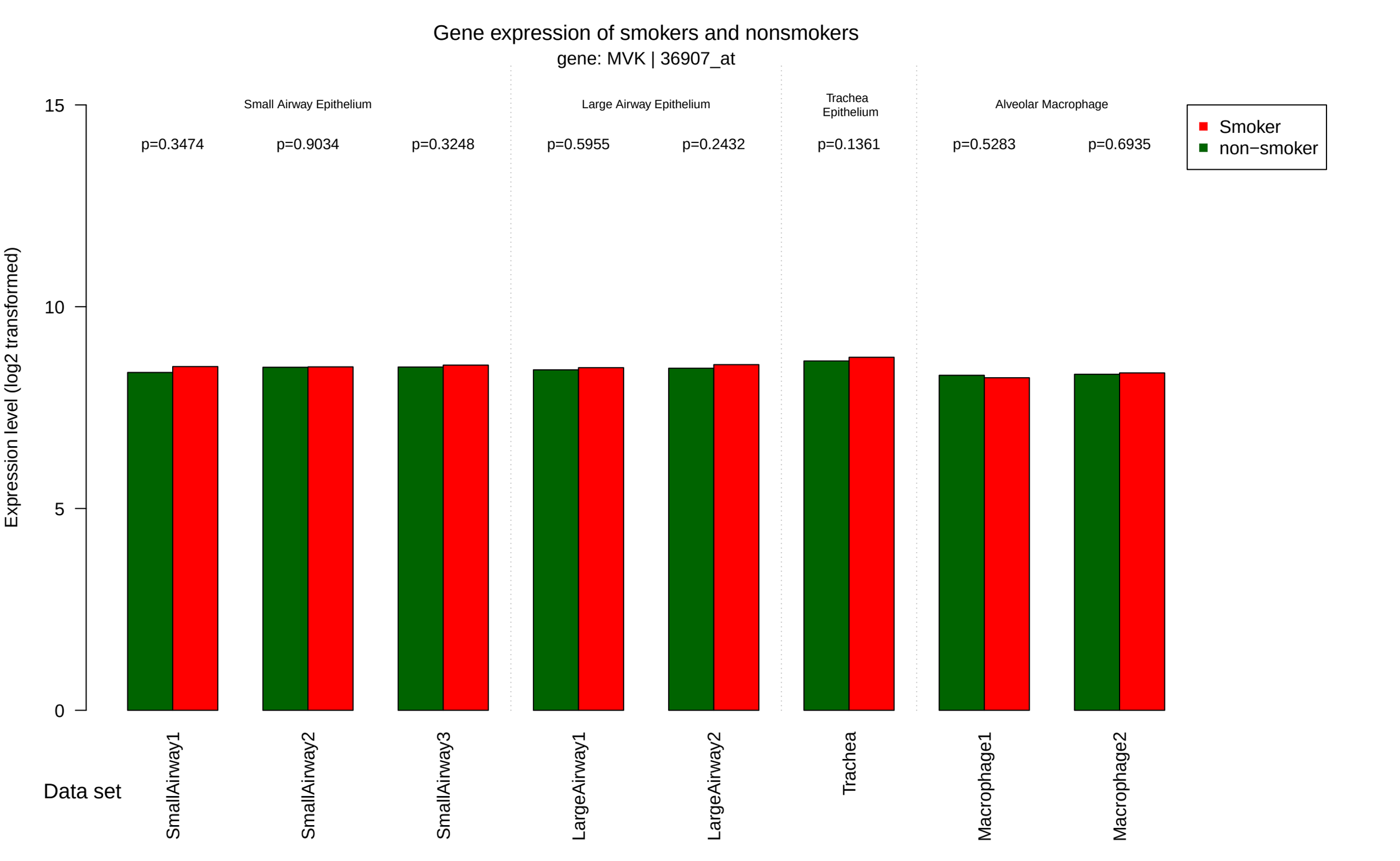
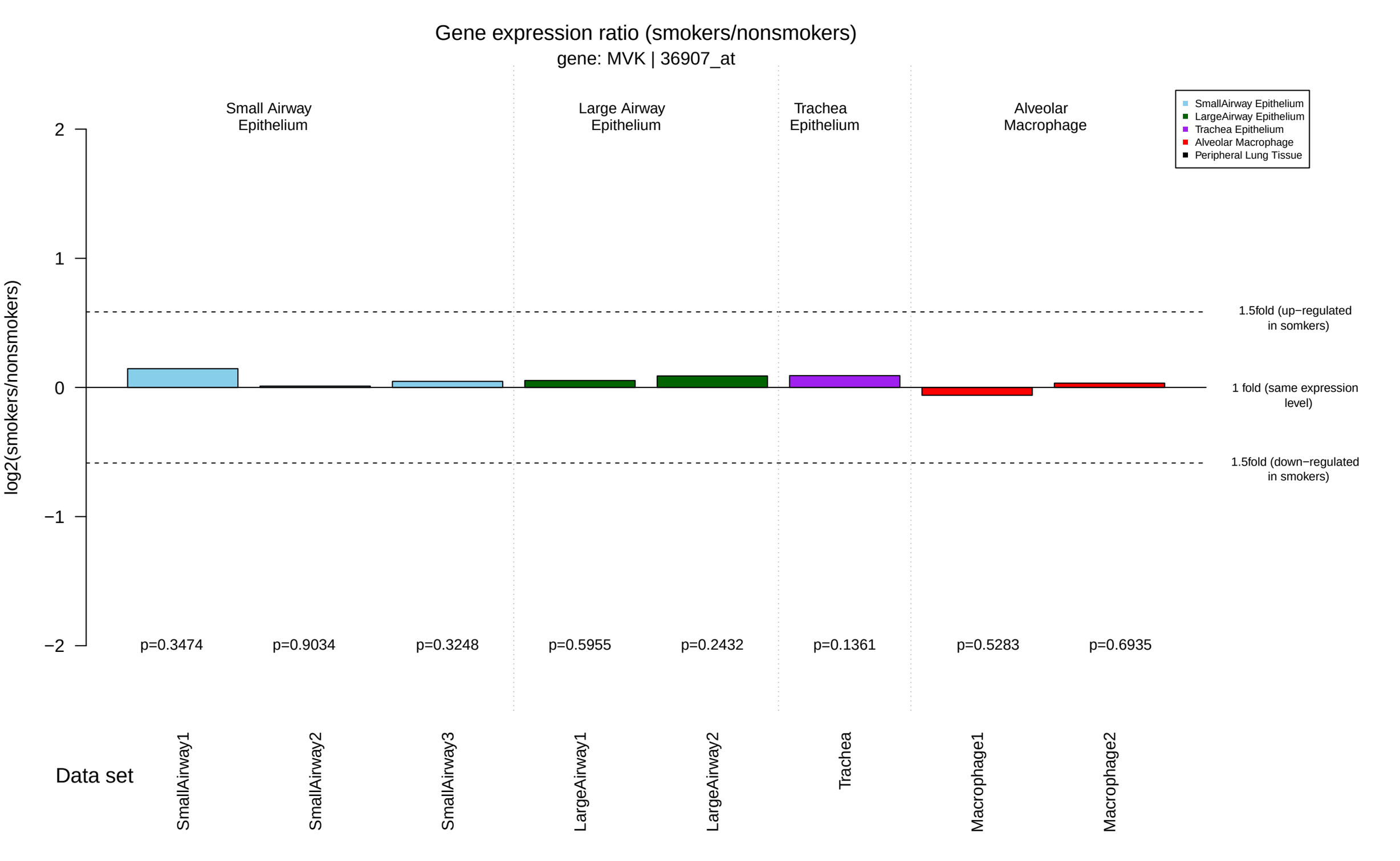
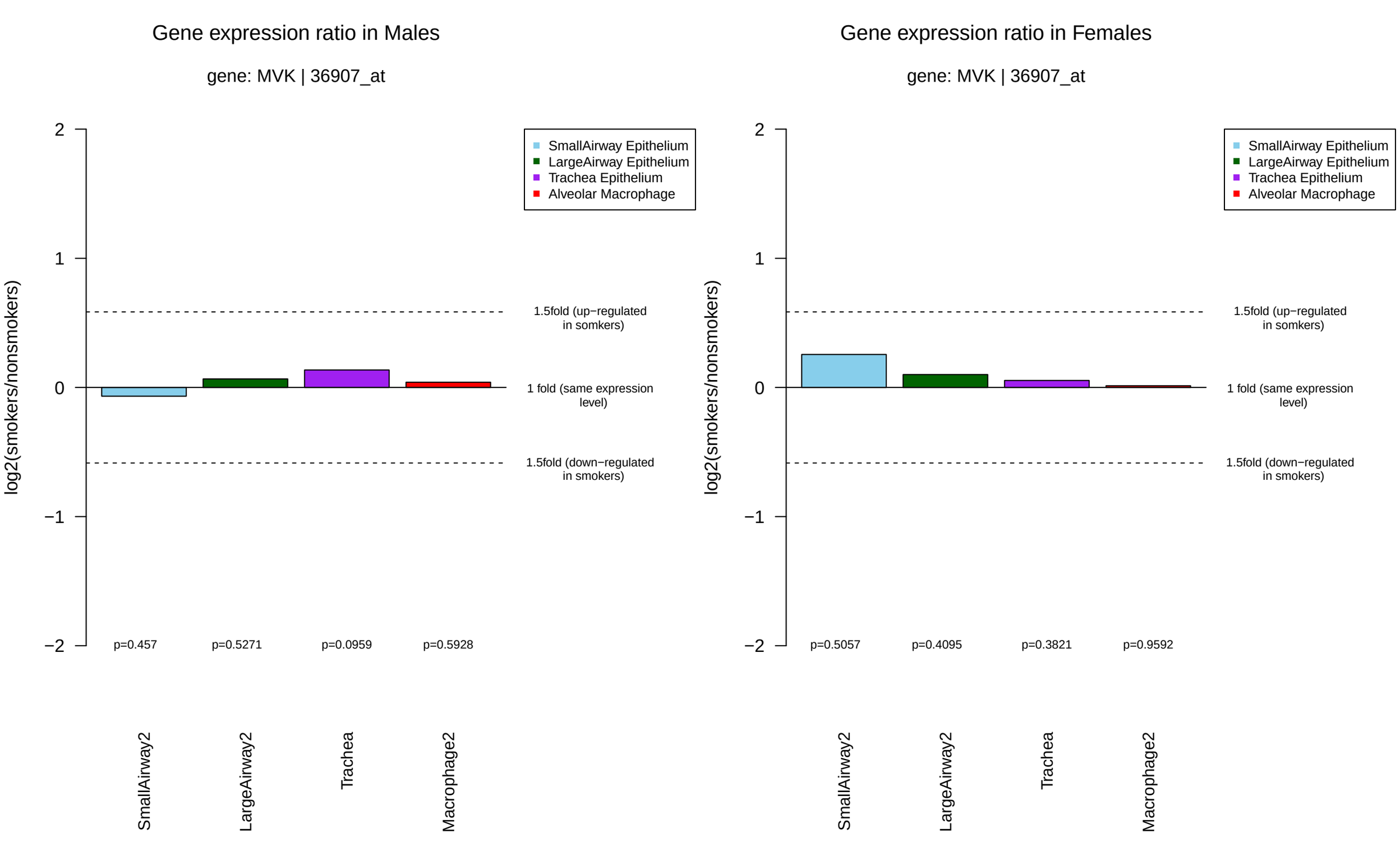
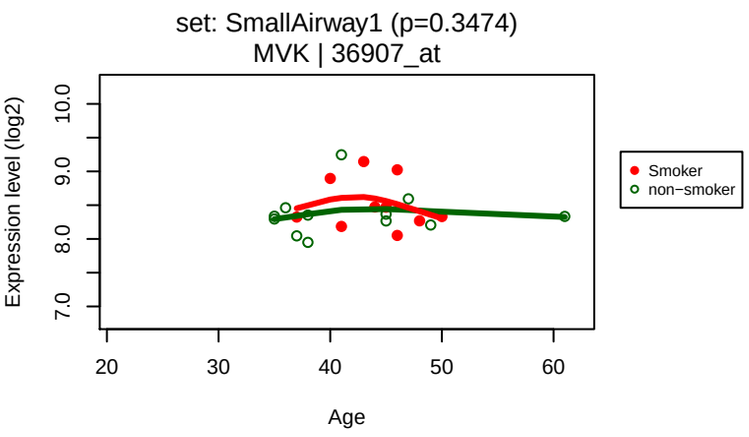
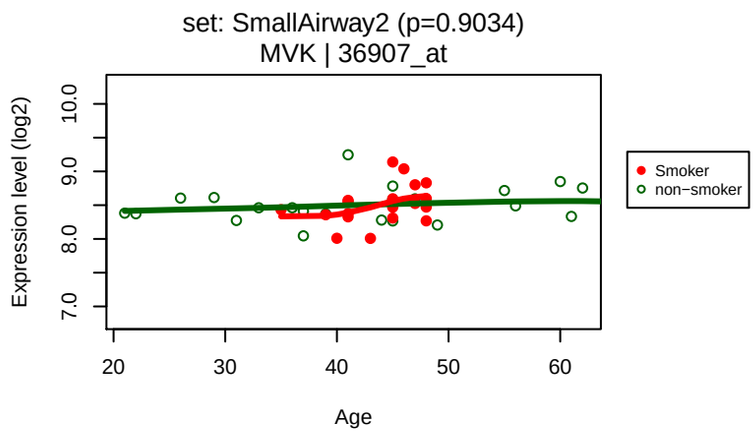
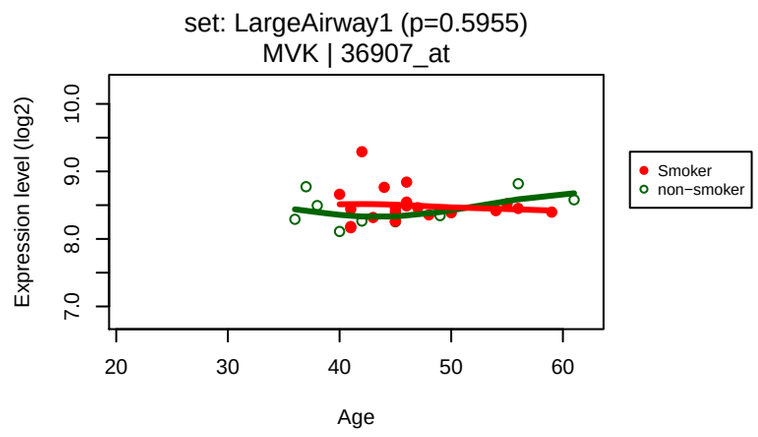

Gene expression of smokers and nonsmokers
gene: MVK | 36907_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3474
p=0.9034
p=0.3248
p=0.5955
p=0.2432
p=0.1361
p=0.5283
p=0.6935
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

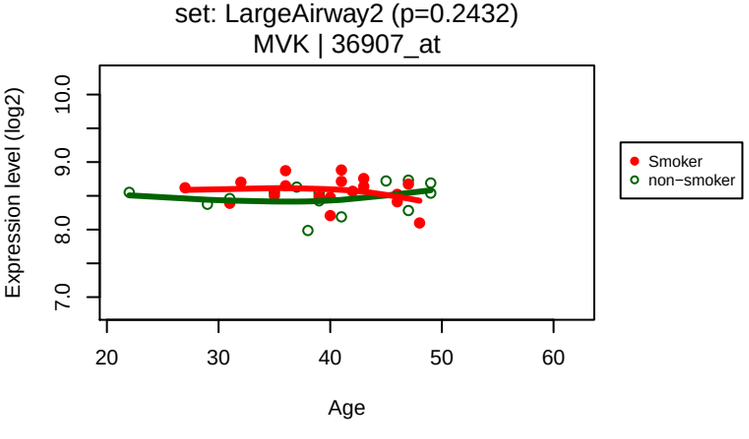

Gene expression ratio (smokers/nonsmokers)
gene: MVK | 36907_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3474
p=0.9034
p=0.3248
p=0.5955
p=0.2432
p=0.1361
p=0.5283
p=0.6935
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

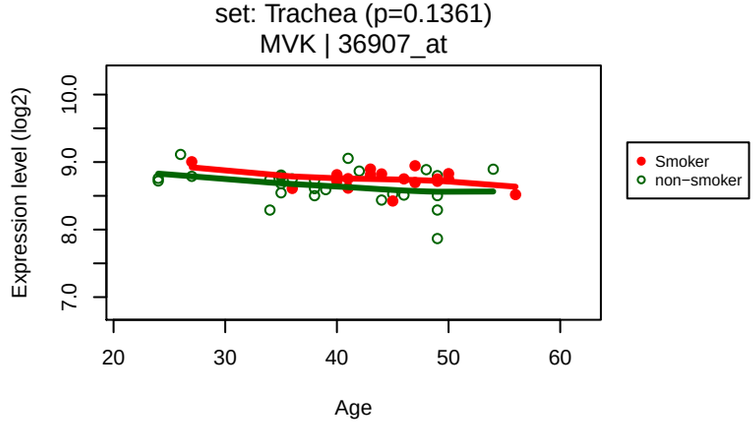

Gene expression ratio in Males
gene: MVK | 36907_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.457
p=0.5271
p=0.0959
p=0.5928
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MVK | 36907_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5057
p=0.4095
p=0.3821
p=0.9592
SmallAirway2
LargeAirway2
Trachea
Macrophage2

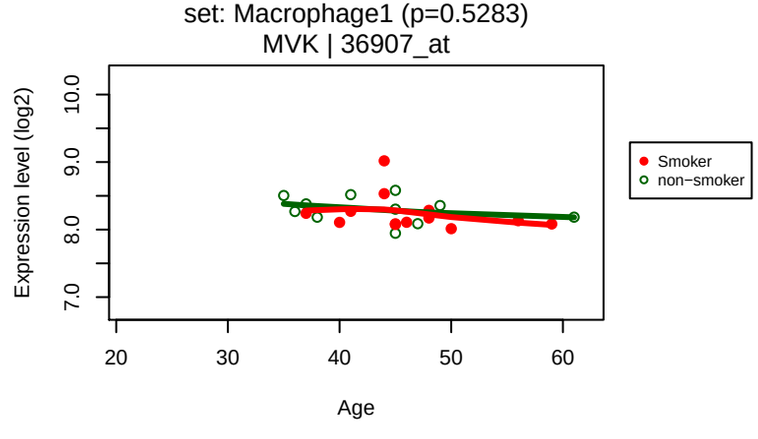

set: SmallAirway1 (p=0.3474)
MVK | 36907_at
Expression level (log2)
7.0
8.0
9.0
10.0
20
30
40
50
60
Age
Smoker
non−smoker

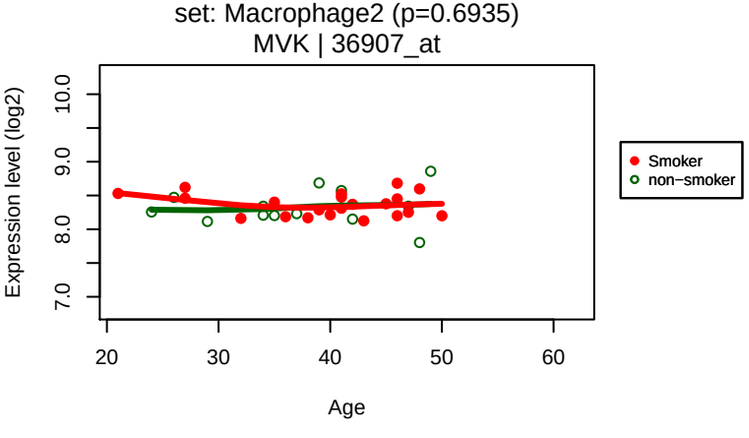

set: SmallAirway2 (p=0.9034)
MVK | 36907_at
Expression level (log2)
7.0
8.0
9.0
10.0
20
30
40
50
60
Age
Smoker
non−smoker

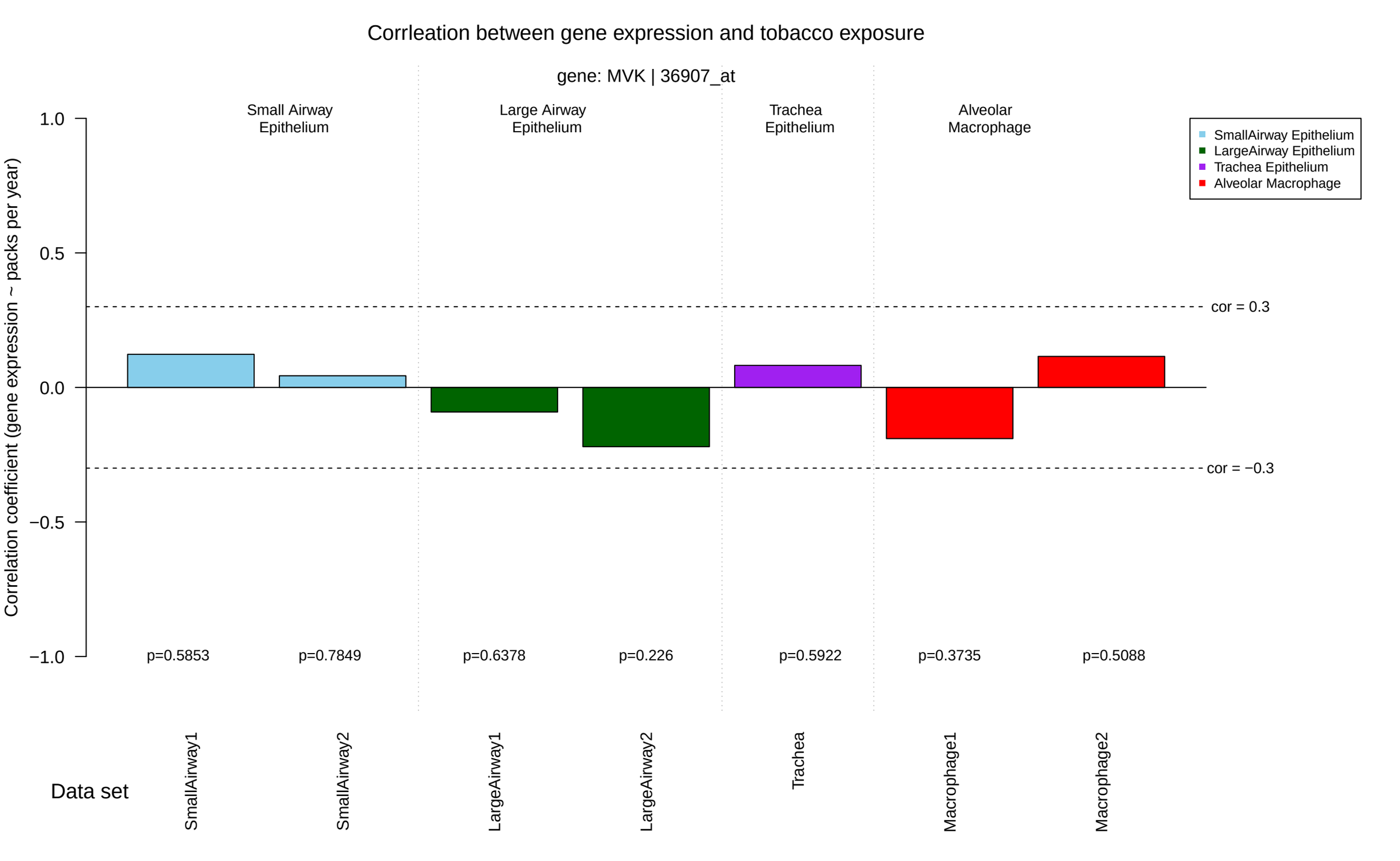

set: LargeAirway1 (p=0.5955)
MVK | 36907_at
Expression level (log2)
7.0
8.0
9.0
10.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2432)
MVK | 36907_at
Expression level (log2)
7.0
8.0
9.0
10.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.1361)
MVK | 36907_at
Expression level (log2)
7.0
8.0
9.0
10.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5283)
MVK | 36907_at
Expression level (log2)
7.0
8.0
9.0
10.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6935)
MVK | 36907_at
Expression level (log2)
7.0
8.0
9.0
10.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MVK | 36907_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5853
p=0.7849
p=0.6378
p=0.226
p=0.5922
p=0.3735
p=0.5088
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2